Sahaj Bhadja

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Summary

Software Engineer with an MS in CS and 2+ years shipping full-stack features across **TypeScript**, **React**, **Node.js**, and **Postgres**, from database queries to API endpoints to UI components, including a hackathon-winning **healthcare** app.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Object-Oriented Programming, Operating Systems, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Designed normalized **Postgres** and Azure SQL schemas with row-level security protecting each tenant's sensitive records for 500+ users, cutting dashboard latency from 1.2s to 350ms through query tuning.
- Built FastAPI endpoints and background workers with explicit job states and failure isolation, processing 1K+ AI jobs at 3x throughput under concurrent load while integrations held 99%+ production **reliability**.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Shipped full-stack features end to end, from **React** upload and grading-history views to Flask REST endpoints and MySQL tables, serving 150+ instructors and 10K+ daily requests in production.
- Debugged production OCR and timeout failures by adding model fallback routing, and built concurrent ingestion for 1K+ files per batch, cutting feedback turnaround 60% via continuous **deployment**.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- **Refactored** large-scale API integrations into modular object-oriented Singleton and Factory designs with caching and pagination, improving response times 30% for low-bandwidth users and reducing technical debt 20% in existing integration code.
- Wrote 20+ unit and integration test suites in nightly Jenkins CI/CD and collaborated closely with Microsoft and GitHub **stakeholders** in Agile sprints from requirement analysis through deployment and production support.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*

- Built eight **TypeScript** and **Node.js** microservices with React for authentication, customer records with interaction history, billing, and notifications, syncing 2M+ records across multiple locations in real time via WebSockets.
- Implemented JWT authentication and role-based access, automated Jest and Selenium tests in Jenkins CI/CD at 95%+ build validation, and used analytics to drive changes that cut wastage 42%.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Built home-services **booking** APIs with time-slot selection, pricing, coupons, and Firebase real-time status updates, using transactional locking to prevent double-booked slots and sustaining sub-200ms latency on core flows.
- Translated client requirements into an end-to-end architecture with 12+ integration points and automated refunds through idempotent payment **webhooks**, cutting manual payment handling by 70% across 50K+ booking records.

Skills

Languages: TypeScript, JavaScript, Python, SQL, Java, C++, HTML, CSS
Frontend: React, Next.js, Shadcn, TailwindCSS, Responsive Design
Backend: Node.js, Express.js, FastAPI, Flask, REST APIs, WebSockets, Webhooks, JWT
Data: PostgreSQL, MySQL, MongoDB, Firebase, Azure SQL, Row-Level Security, Query Optimization
Delivery: Git, GitHub Actions, CI/CD, Jenkins, Docker, AWS, Vercel, Sentry, Jest, Selenium
Fundamentals: Data Structures, Algorithms, Software Design, Debugging, Code Reviews

Projects

HMIS – Hospital Management Information System Live | *TypeScript, React, TailwindCSS, MERN, Twilio, Sentry*

- Won Smart India Hackathon with a TypeScript and **React** healthcare app giving doctors consent-based access to patient histories, appointment booking, and Twilio reminders, cutting administrative workload 30% across 100+ daily appointments.

Big-Brain Live | *React, Shadcn, Convex, Clerk, PostgreSQL, LLaMA-2*

- Built and shipped a collaborative document app with a responsive **Shadcn** UI, Clerk-secured access, Convex real-time sync, and LLaMA-2 summaries, cutting document review time 60% for 200+ users.

FinNexus Live | *Next.js, TypeScript, TailwindCSS, MongoDB, Kubernetes*

- Built a **TypeScript** and Next.js platform with TailwindCSS components, Plaid bank linking, a token-based firewall, and real-time WebSocket updates, deployed via CI/CD for 50K+ users across 1M+ transactions.